It is important to capture the variation that does exist from previous research as it relates to Spanish vowels. Below is a review of Spanish vowel variation that is found in L1 (first language speakers) of Spanish around the Spanish speaking world.

1. Variation in the Spanish Vowel System

This review will outline regional and sociolinguistic variation of Spanish vowel sounds previously covered in empirical research. Later, a summary of the scope and nature of this variation is presented.

1.1. Regional Variation

Previous empirical works have challenged the notion that the Spanish-speaking world realizes its vowel sounds in a uniformly symmetrical way. For clarity, I will shortly define tonicity here. Tonicity is the placement of the main stress in a word. A stressed syllable is called a tonic syllable. Syllables before it are pre-tonic, and those after it are post-tonic. For example, in *camino* (I walk) [kaˈmino], mi is tonic, ka is pre-tonic, and no is post-tonic.

There are differences that have been found in how Spanish vowels are distributed and how long they are across different dialects. To begin, Quilis and Esgueva (1983) analyzed Mexican, Ecuadorian, Chilean, and Peninsular Spanish vowels in twenty-two speakers. They found that Latin American speaker's vowels were slightly longer than Peninsular speaker's vowels with no consistent patterns of vowel-quality differences. Morrison and Escudero (2007) likewise observed that Peninsular vowels were shorter relative to Peruvian vowels when comparing speech from seventeen speakers from Lima, Peru and Madrid, Spain. They found that the only differences were F2 measurements of the phoneme /o/. Morrison and Escudero (2007) concluded that vowel quality remained largely stable between Peruvian and Peninsular Spanish. Still, there is regional variation to be seen within Peruvian Spanish. For example, speakers from Cuzco tend to produce vowels with a wider vowel space and more fronting compared to speakers from Lima (O'Rourke, 2010). Chládková, Escudero, and Boersma's (2011) analysis comparing speakers from Madrid, Spain and Lima, Peru revealed that Peruvian vowels were realized more peripherally, longer, and had higher F0 than Peninsular vowels. Santiago and Mairano (2018) showed that stressed vowels are longer than unstressed ones in both Peninsular and Mexican Spanish and that Mexican speakers have longer vowel sounds overall. In both varieties, [a] and [o] were more prone to reduction in unstressed contexts than [i, e, u]. Marchini and Ramsammy (2022) determined that mid and low vowels in unstressed closed syllables (syllable ending in a consonant) were produced more centrally and shorter in CVC environments in Altiplateau Mexican Spanish speakers.

Spanish vowel variation is also captured by changes that occur as a process of interactions with adjacent segments (consonants or vowels) (Martínez Celdrán & Elvira-García, 2019). In eastern Andalusian Spanish, word-final /s/ weakening can trigger vowel changes that preserve morphological distinctions. For example, /e/ and /o/ can be realized as open [ɛ] and [ɔ] before weakened /s/ in order to communicate final consonant loss of a plural noun (Hualde, 2005). Herrero de Haro (2017, 2022) studied /s/ aspiration, among other variations, in eastern Andalusian Spanish and discovered that adjacent consonantal sounds to mid vowels led to vowel lowering and greater openness in the vowel space.

A variation process called mid-vowel raising (/e/ → [i]; /o/ → [u]) has also been accounted for as a process of systematic vowel change in Spain, Mexico, and Puerto Rico. As an example, vowel raising was found in a northwestern village in Spain where /o/ was realized as [u] in word final contexts (e.g., *hermano* [eɾˈmano] brother pronounced as [eɾˈmanu]) (Holmquist, 1985). Another example of vowel raising with /e/ would happen if a speaker pronounces *leche* (milk) as [ˈlet͡ʃi]. Holmquist (1998, 2005) reported that mid-vowel raising was frequent in word-final contexts after high-vowel glides or palatal consonants among rural Puerto Rican speakers.[1] Oliver Rajan (2007) likewise observed that mid-vowel raising in Puerto Rican Spanish was more common after palatal consonants. Barajas (2014) conducted an acoustic study of rural Michoacán Spanish and showed that /e, o/ raising was more frequent after palatal consonants and with preceding high vowels and that /e/ raising to [i] happened more often than /o/ raising to [u].

[1] i.e. palatal nasal /ɲ/ in *año* (year), voiced palatal approximant /ʝ/ *llanta* (flame-pronunciation from yeísmo (sh sound for */ll/*), voiced palatal lateral approximant /ʎ/ *llanta* (flame-non-yeísmo, y-sound), voiceless postalveolar affricate /tʃ/ in *chica* (girl).

Cross-dialectal variation of Spanish vowels can also be found in unstressed vowel devoicing in Mexican, Bolivian, Ecuadorian, Peruvian, and Colombian vowel systems. This process is often referred to as unstressed vowel reduction (UVR) and it occurs when a vowel is either partially or fully devoiced from pronunciation. For example, unstressed vowel devoicing can occur when /e/ in, *un caf(e)cit*o (a little coffee), is almost soundless or completely soundless in its production. In Mexico City Spanish, unstressed mid-vowel weakening, especially /e/, has been observed in post-tonic positions, before voiceless consonants (mostly /s/), and in word-final contexts (Lope Blanch, 1964; Serrano, 2006).[2] Dabowski (2020) found that unstressed vowel weakening in Mexico City Spanish favored preceding voiceless contexts in post-tonic syllables.

Similar patterns have surfaced in other varieties. Delforge (2008) found that /e, i, u/ in Cuzco Spanish were devoiced in unstressed, post-tonic, word-final positions that were also adjacent to voiceless consonants, especially /s/, and this paralleled findings from Mexico City Spanish. Lipski (1990) showed that /e/ and /i/ in unstressed, post-tonic positions before /s/ underwent devoicing and centralization in Ecuadorian Spanish. Sessarego (2012a) analyzed Bolivian Cochabambino Spanish and found a continuum from partial devoicing to full deletion that was most prominent in /o, i, a/ in post-tonic, word-final positions that were before voiceless consonants, again, mainly before /s/. Similar contexts were found to cause devoicing of /e/ and /o/ in Afro-Yungeño Spanish (from Bolivia's Yungas region) (Sessarego, 2012b). Correa and Rodríguez (2024) analyzed speech from speakers of Bogotá Spanish and reported that 7% of vowels in their study were devoiced in 97% of unstressed positions following /s/.

Some scholars criticized the term "reduction" for the phenomenon of unstressed vowel weakening because vowel realizations are not always centralized or shifted in quality during this process. They exhibit less voicing of the vocal folds which in turn make them less visible in spectrograms and perceptually harder to hear. As Delforge (2008) argues, "Since the 'reduced' vowels appear to exhibit degrees of devoicing rather than quality changes… as opposed to centralization… the phenomenon is

[2] Voiceless consonants are those produced without vibration of the vocal cords. The Spanish voiceless consonants are /p, t, k, f, θ, s, x, tʃ/.

more accurately described as devoicing" (p. 121). Delforge's view places emphasis on the fact that devoicing variation doesn't necessarily cause changes to vowel quality or distribution.

Collectively, previous research has highlighted that Spanish vowel variation manifests in several regional differences, including vowel duration and distribution, unstressed vowel devoicing/weakening, coarticulatory effects (namely with /s/ aspiration), and vowel raising. Most of the variation in Spanish vowels has been linked to regional differences in quantity, namely durational differences. The main processes that have been found to cause distributional changes in vowels, and consequently the sound quality of these vowels, are co-articulatory effects and vowel raising.

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